

Group Project: Designing a Data Architecture

Overview

Through this group project, you will apply the concepts covered in class to design a complete, end-to-end data architecture for a domain of your choice. Groups are 3 students each.

Learning Goals

Through completing this project, make sure you can:

- translate a real-world domain into a structured data model
- Make and defend schema design decisions (star vs. snowflake, OLAP vs. OLTP)
- Design and implement at least one cloud ETL process
- Orchestrate your pipeline using Airflow
- Communicate your design clearly to a technical audience

Picking a domain:

Business goal:

Your architecture exists to support analytical and “business” use cases. You must identify a broad goal and 3-4 specific analytical use cases relevant to that goal that can be answered using data.  Examples from last year

Technical depth:

Pick a domain that satisfies 1-3 of these characteristics:

- Multi-source or multi-format data
 - Different sources, data providers, or APIs, etc.
 - structured tables, semi-structured files (JSON, Parquet), and unstructured content (text documents, logs, images, or video)
- Highly scalable (100 GB to TBs)
 - Either handle such a large dataset or be ready to make and defend decisions that will work at that scale.
- Governance and Security
 - Highly sensitive domains — healthcare, finance, legal, and education.
 - Governance constraints drive your architecture decisions – access control, auditing, anonymization, etc.
- Temporal Richness
 - Data changes and accumulates meaningfully over time – and you must show this in your project.
 - Example: event streams, daily transactions, sensor readings, etc.

Sources: You are encouraged to use publicly available datasets, but in many domains, you may not find public datasets that satisfy all your constraints. **Start from a real dataset, but be prepared to augment it with synthetic data.**

Milestones

Milestone (total 100 pts)	Details	Submission
M1 — Domain & Dataset (10 pts)	Align on a domain and identify your data sources. Include: team name (after a food item), member names, chosen domain, analytical question(s), and dataset sources.	Upload your Google Slides link to Canvas. Use this same deck for all milestones — do not create a new deck.
M2 — Domain Model (10 pts)	Apply domain modeling to your domain. Include an ER diagram with at least 5–6 entities and relationship annotations. Identify your data needs from the ER diagram and compare them with your data sources. Make an initial guess about which entities become fact vs dimensions	No separate submission. Add slides to your existing deck – will review in class.
M3 — Midterm Design Pitch (20 pts)	~7 min group presentation. See the Midterm design pitch tab for details.	No separate submission. (hide any slides you are not presenting, but keep everything in one deck)
M4 — GitHub Repo (10 pts)	Submit your GitHub repo link + checklist of everything that's on GitHub.	No separate submission - add a slide to your deck on how I can access the GitHub repo + checklist
Final Presentation & Demo (50 pts)	~10 min group presentation with a working demo. See the Final presentation tab for details.	No separate submission. (hide anything you are not presenting, do not delete)

Also see the separate tab on the “Bonus” component (+20).

A Note on the Lakehouse Course

For MSDS 681 (Data Lakehouse) next term, this architecture project will serve as the design foundation for the lakehouse you build there. You are not required to continue the same project, but starting with a solid architecture design will be a real advantage.